

Project Planning and Management

1. Project Proposal

Project Title:

Student Performance Analytics Dashboard: Evaluating Educational Outcomes

Project Overview:

This project aims to design and develop a student performance analytics dashboard to evaluate the impact of learning on educational outcomes. The system is built to process, organize, and analyze educational data collected from students through pre-test and post-test assessments, enjoyment scales, and passion measurement instruments. The project combines data engineering, database design, statistical analysis, and visualization techniques to provide meaningful insights that support educational decision-making.

Project Objectives:

- Design a structured and secure data management system for student performance data.
- Build a star schema database model that supports analytical processing and reporting.
- Measure students' knowledge gain by comparing pre-test and post-test results.

- Analyze enjoyment and passion indicators to understand their relationship with learning outcomes.
- Develop an interactive dashboard for visualizing key student performance metrics.
- Apply data anonymization and basic security practices to protect student privacy.

Project Scope:

In Scope:

- Data cleaning and preprocessing of the available educational dataset.
- Database design, including ERD and star schema implementation.
- Development of SQL-based structured tables for dimensions and fact tables.
- Statistical analysis of student performance indicators.
- Dashboard design and visualization of key findings.
- Documentation and presentation of project results.

Out of Scope:

- Real-time integration with Moodle or other learning management systems.
- Mobile application development.
- Deployment on cloud-based production servers.

2. Project Plan:

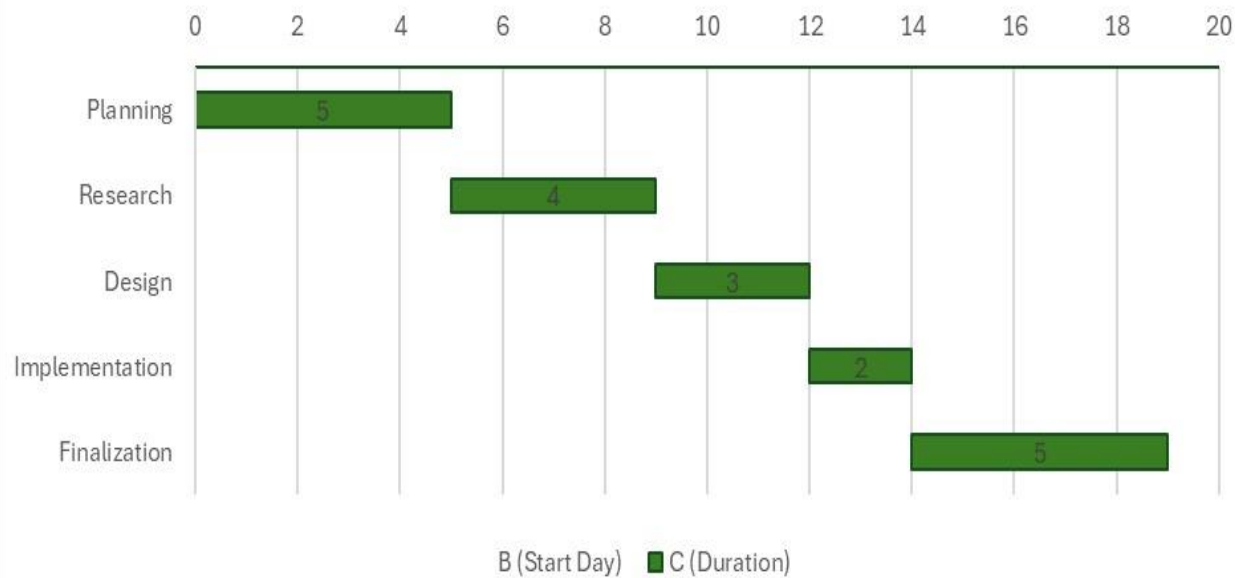
Timeline and Milestones

Phase	Main Activities	Target Completion
Planning	Project proposal, management plan, scope definition	5 May 2026
Research	Literature review and requirements gathering	9 May 2026
Design	ERD, star schema, diagrams, and architecture	12 May 2026
Implementation	SQL database, analysis scripts, dashboard draft	14 May 2026
Finalization	Final review, GitHub upload, and submission	19 May 2026

Deliverables

Deliverable	Description	Due Date
Project Planning & Management	Proposal, plan, risks, KPIs, and responsibilities	5 May 2026
Literature Review	Summary of related research and tools	9 May 2026
Requirements Gathering	Stakeholders, use cases, and requirements	11 May 2026
System Analysis & Design	ERD, DFD, diagrams, and architecture	12 May 2026
Database Implementation	SQL star schema and structured tables	14 May 2026
Dashboard Prototype	Visual presentation of student performance	15 May 2026
Final Submission	GitHub repository and supporting files	19 May 2026

Project Timeline - Student Performance Dashboard



Resource Allocation

Resource	Purpose
Python	Data cleaning, preprocessing, and statistical analysis
SQLite / SQL	Database design and implementation
Power BI / Python charts	Data visualization and dashboard creation
GitHub	Version control and project submission
Microsoft Word	Documentation and report preparation

3. Task Assignment & Roles:

Role	Responsibilities
Project Manager/ Alshiamaa Fouad Mohammed Kamal	Planning, coordination, follow-up, and submission management
Data Engineer/ Dina Fouad Esmail Fath El-bab	Data cleaning, preprocessing, and database preparation
Data Engineer/ Eman Fadel Kamal El-Din	Star schema implementation and SQL table creation
Data Engineer / Alshaimaa Fouad Mohamed Kamal	Statistical analysis and interpretation of results
Data Engineer/ Hend Mohammed Hasanein Abo-El-Naga	Visualization design and dashboard development
Data Engineer/ Eslam Mohamed Hussien Ali	Writing reports, organizing deliverables, and final formatting

4. Risk Assessment & Mitigation Plan:

Risk	Impact	Mitigation Strategy
Data quality issues	High	Apply validation rules, cleaning scripts, and data review
Missing or inconsistent values	Medium	Use imputation and standardization methods
Time constraints	High	Follow a strict daily plan and prioritize core deliverables
Data privacy concerns	High	Anonymize student identifiers and avoid exposing sensitive data
Incomplete documentation	Medium	Finalize each section immediately after completing the related task

5. Key Performance Indicators (KPIs):

KPI	Measurement Target
Data cleaning completion rate	100 percent of records reviewed and processed
Database readiness	All dimension and fact tables created successfully
Query accuracy	Correct SQL output for all main analytical queries
Dashboard readiness	A functional prototype completed before submission
Documentation completion	All required sections completed and uploaded
Submission readiness	All files uploaded to GitHub before 20 May 2026